Model-specific focal points in one-shot visual coordination

Steps 1 and 1b — GPT-5.4 and Claude Opus 4.8

LLM Coordination project · report generated 2026-08-06 · all figures and statistics computed from the run CSVs at build time

Summary

- **Both models coordinate on visual form, and they do it differently.** Across a five-pair battery and a 30-pair screen, dyadic match probability ranges from chance to 1.000 depending on the stimulus, and the two models dissociate on 8 of 30 screened pairs.
- **Crossovers exist in both directions.** 5 pairs where GPT is more concentrated, 3 where Claude is. The strongest single result is `s_arrow` (Claude 1.000 vs GPT 0.503).
- **A model that cannot find a form rule does not answer randomly — it defaults to a side, and then justifies that choice in form language.** This is the central methodological finding and it invalidates any analysis that reads a 50/50 identity split as “no preference” without checking position.
- **Competing-cue stimuli did not produce dissociations.** 18 of 30 pairs sat at a perfect 1.000/1.000 ceiling, including 9 of the 14 pairs built specifically to put two cues in conflict. Both models resolve cue conflicts the same way.

1. Task and procedure

Each trial is an independent, one-shot API call with no conversation history. The model sees a single composite image containing two shapes side by side and must pick the one it believes another instance of itself would also pick. There is no communication and no feedback.

Prompt

Sent as a single user text block alongside the image, with no system prompt. **No labels — A/B or otherwise — are shown to the model.** The A/B identity coding exists only in the runner and the analysis.

> You are playing a one-shot coordination game with ONE other player - another instance of you (the same AI model). You will each independently pick one of the two shapes shown; you want to pick the SAME shape as the other player. No communication. IMPORTANT: the left/right position of the two shapes is randomized independently for each of you, so position CANNOT be used to coordinate - you must coordinate on a shape by its FORM. Give one short sentence of reasoning, then end with exactly 'CHOICE: LEFT' or 'CHOICE: RIGHT' (the position in YOUR image of the shape you chose).

Hash `3003ae1b`, stamped on every row of every run reported here. Byte-identical to the arm-1 prompt of the project’s earlier `confidence_calibration_probe.py`, so this data is comparable with that series.

Counterbalancing and recoding

Each pair exists in two layouts: `AB` places identity 1 on the left, `BA` places identity 2 there. Allocation is exactly half and half in every cell. The model answers with a *position* (`CHOICE: LEFT` / `RIGHT`), which is recoded offline to a stimulus identity via the layout. Because the counterbalance is exactly balanced, a pure position strategy yields a 50/50 identity split — it reads as chance rather than as a spurious preference.

Models and settings

	GPT	Claude
model id	`gpt-5.4`	`claude-opus-4-8`
endpoint	Responses API	Messages API
reasoning / thinking	effort low, summary auto	adaptive, summarized
max tokens	not set	2000
temperature / top_p / top_k	not set	not set
seed	not exposed by the API	not exposed by the API

Settings were fixed across every call; each model shows exactly one distinct settings string in the data. The only seeded component is the counterbalance RNG (`20260804`), which is deterministic.

2. Measures

Unbiased dyadic match probability. With counts n_k over identities and N trials, $\Sigma_k\, n_k(n_k-1) \,/\, N(N-1)$. This is the probability that two *distinct* trials agree — the right quantity for "would two independent instances pick the same shape". The plug-in Σp_k^2 is reported alongside as a check and runs slightly higher. Chance is 0.5. Confidence intervals are percentile 95% from 5,000 bootstrap resamples at a fixed per-cell seed.

Conditional-position diagnostic. For a model's modal identity m, compute the rate at which m is chosen separately under each layout: $p(m \mid m$ on the left) and $p(m \mid m$ on the right). A model choosing by **form** picks m at the same rate either way. A model choosing by **side** picks m only when it happens to sit on the favoured side, driving the two rates to 1 and 0. Reported as *form consistency* $= 1 - |p_{AB} - p_{BA}|$.

> **This diagnostic is necessary but not sufficient.**
> A model answering at random also scores near 1 on form consistency, so it is only meaningful read jointly with a match rate above chance. Both conditions are enforced wherever it is used as a criterion.

3. Step 1 — five-pair battery

600 calls (5 pairs × 2 models × 60), 405 s, zero unparseable responses, zero API errors, 60/60 valid in all ten cells.

pair	GPT	95% CI	P(left)	Claude	95% CI	P(left)
`proto`	1.000	[1.000, 1.000]	0.50	1.000	[1.000, 1.000]	0.50
`inkblot`	0.967	[0.903, 1.000]	0.52	0.491	[0.491, 0.537]	0.00
`rot`	0.696	[0.587, 0.817]	0.45	0.741	[0.619, 0.873]	0.45
`mirror`	0.573	[0.506, 0.696]	0.77	0.497	[0.491, 0.548]	0.85
`noise`	0.506	[0.491, 0.573]	0.38	0.494	[0.491, 0.548]	0.97

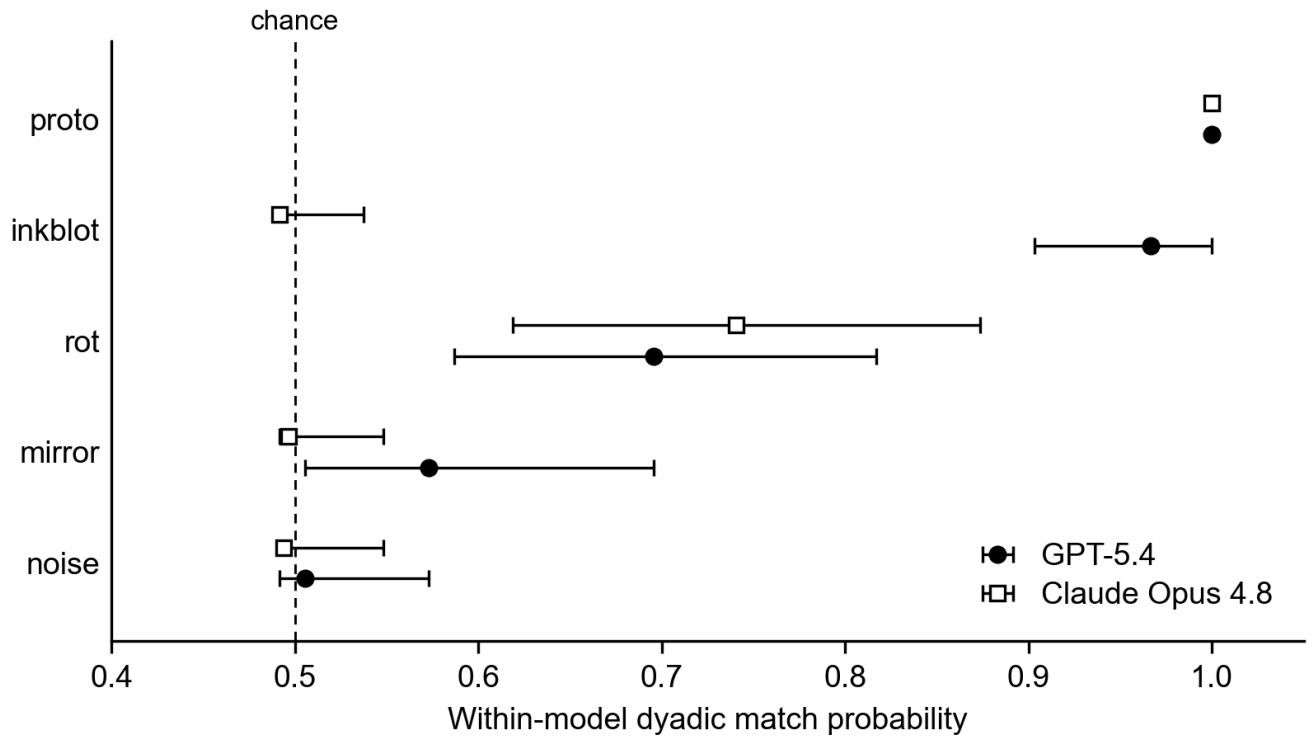


Figure 1. Within-model dyadic match probability by stimulus. Filled circles GPT-5.4, open squares Claude Opus 4.8. Bars are 95% bootstrap CIs (5,000 resamples). Dashed line marks chance.

Three distinct mechanisms, not one

Requiring a sentence of reasoning before the choice made the failure modes legible, and they turned out to be different in kind.

1. **Genuine form coordination** — `proto` (both models 1.000) and `rot` (0.695 / 0.741). On `rot` the chosen side flips with the layout, which is the signature of tracking the shape rather than the position.
2. **Honest fallback** — Claude on `noise`, which states the position openly: "These both appear to be random noise with no clear form, so I'll default to a simple tiebreaker." There is no form signal to find; the model says so and picks a side deliberately. The measured 0.494 is the correct answer, not an artifact.
3. **Confabulated side default** — Claude on `inkblot`, answering `RIGHT` on 60 of 60 trials while giving a *form* justification each time, accurately describing whichever shape occupied the right slot: [layout BA] "The symmetric, star-like shape with balanced points is a more natural Schelling focal point than the irregular blob." [layout AB] "The right shape has two distinctive tall pointed spikes at the top, making it the more memorable, easily-agreed-upon form." Different features cited depending on what sits on the right, and whichever it is gets called the focal point. The choice tracks position; the justification is fitted to it afterward. Identity lands at exactly 30/30.

Why `mirror` fails where `rot` succeeds

This is a property of the stimuli, not of the models. `rot` is a blob rotated 90°, so one option is upright and one is sideways — a position-independent asymmetry that *both instances share a prior about*. `mirror` is a blob reflected across the vertical axis: reflection preserves area, complexity and lobe count, leaving only handedness, and handedness is exactly the dimension with no shared prior to break the tie. A rule can be stated, but there is no reason to prefer one sign of it, so rule-selection is itself a coin flip. Claude generated two incompatible chirality rules minutes apart:

"…coordinate on the shape with its lower point angling toward the **right**."
"…pick the one whose narrow point tucks toward the **bottom-left**."

> **The modal share is the misleading statistic on `mirror`, not the match rate.**
> GPT's apparent "70% preference for blob-mirror" comes almost entirely from the BA layout, where blob-mirror sits on the left: it was chosen 29/30 there but only 13/30 when the same shape sat on the right. The match rate is unaffected — under a balanced counterbalance a side bias can only push it toward chance, never above it — so 0.573 remains an honest, conservative estimate.

4. Step 1b — 30-pair screen

1,800 calls (30 pairs × 2 models × 30), 862 s, zero unparseable responses, zero API errors. Two families were built:

- **16 single-cue pairs** vary one property with everything else matched — symmetry, complexity, regularity, curvature, convexity, elongation, size, tilt, closure, hole, numerosity, density, alignment, line orientation, contrast, semantic orientation.
- **14 competing-cue pairs** put two cues in opposition (small-symmetric vs large-asymmetric, spiky-small vs smooth-large, and so on), on the hypothesis that a model weighting one cue would diverge from a model weighting the other.

> **The competing-cue hypothesis was wrong.**
> 18 of 30 pairs sat at a perfect 1.000/1.000 ceiling, including 9 of the 14 competing-cue pairs: the models resolve cue conflicts *identically* (size appears to dominate). Five of the eight dissociations came from single-cue pairs. Building conflict into a stimulus did not create disagreement, it created a harder problem that both models solved the same way.

A selection criterion that had to be changed

The pre-specified criteria were: (1) P(left) between 0.35 and 0.65 for *both* models, (2) |Δ| > 0.15 in match probability, (3) form-based choosing. Applied as written, only two pairs passed, both in the same direction — **no crossover**.

The reason is mechanical, and it is the Step 1 finding resurfacing: a model with no form rule side-defaults rather than answering randomly, so a large |Δ| almost guarantees that the *low* model has an extreme P(left). Screening on both models therefore rejects precisely the dissociations being sought. Applying the position criterion to the **concentrated model only** costs nothing: the two models are separate API calls in separate cells, so the low model's side default cannot bias the high model's estimate, and under a balanced counterbalance a side default reads as chance on identity — the correct answer for "no form preference". With that correction, 8 pairs pass and crossovers appear in both directions.

pair	family	GPT	P(left)	form cons.	Claude	P(left)	form cons.	Δ	95% CI on Δ
`s_arrow`	single	0.503	0.23	0.47	1.000	0.50	1.00	-0.497	[-0.517, -0.370]
`c_open_cplx`	compete	0.595	0.30	0.60	0.871	0.43	0.87	-0.276	[-0.460, -0.044]
`s_convexity`	single	0.669	0.30	0.60	0.933	0.47	0.93	-0.264	[-0.434, +0.000]
`s_density`	single	1.000	0.50	1.00	0.814	0.53	0.93	+0.186	[+0.000, +0.331]
`c_star_circle`	compete	0.761	0.50	1.00	0.492	0.07	0.13	+0.269	[+0.074, +0.448]
`c_elong_sym`	compete	0.814	0.47	0.93	0.485	0.03	0.07	+0.329	[+0.129, +0.497]
`s_contrast`	single	1.000	0.50	1.00	0.630	0.60	0.80	+0.370	[+0.186, +0.497]
`s_complexity`	single	1.000	0.50	1.00	0.485	0.10	0.20	+0.515	[+0.434, +0.517]

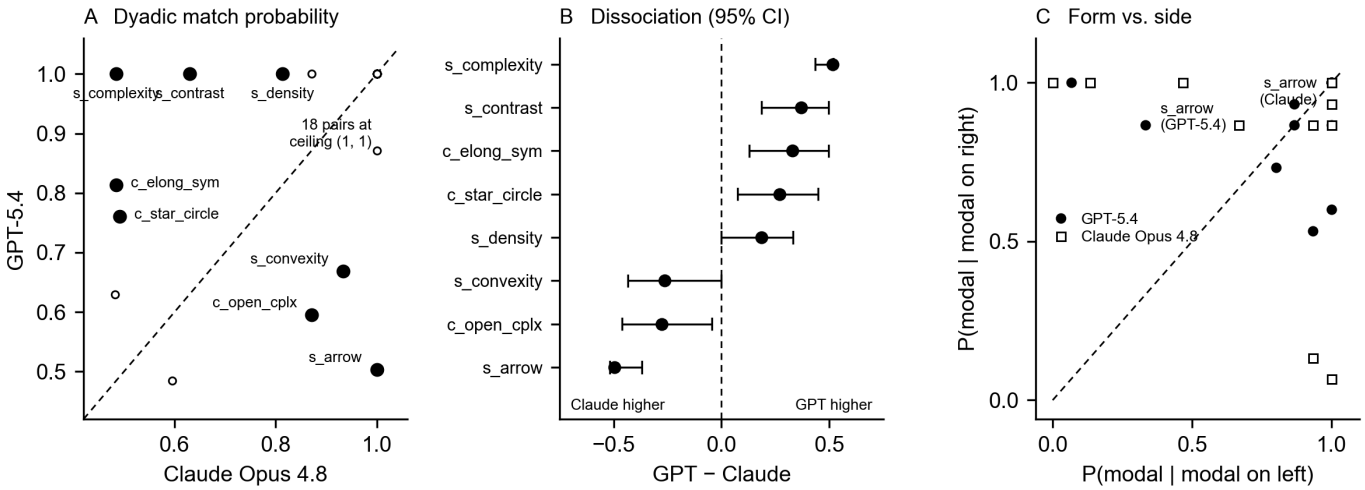


Figure 2. (A) Dyadic match probability, all 30 pairs; distance from the diagonal is the dissociation. Filled points passed all criteria. **(B)** Effect sizes with 95% bootstrap CIs for the selected pairs, both directions. **(C)** Conditional-position diagnostic: rate of choosing the modal identity when it sits on the left versus the right. Points on the diagonal indicate choosing by form; points off it indicate choosing by side. The two labelled `s_arrow` points show the mechanism — Claude at (1.0, 1.0), GPT far off the diagonal.

The strongest result

`s_arrow` (up-arrow vs right-arrow) is the cleanest dissociation in the battery and the Claude > GPT anchor the earlier work lacked. Both models articulate the *same* rule — "up is canonical". Claude applies it on 30 of 30 trials with P(left) exactly 0.50. GPT states it and then does not follow it:

> "The upward-pointing arrow seems the more canonical and salient shape to coordinate on. **CHOICE: RIGHT**" (up was on the right — correct)
>
> "The right-pointing arrow feels like the more standard, salient arrow form. **CHOICE: RIGHT**" (rule switched to fit the side)

GPT answered `RIGHT` on 23 of 30 trials. This is the same confabulated side default Claude showed on `inkblot` in Step 1, with the models' roles reversed.

5. Recommendation for the next experiment

pair	direction	GPT	Claude	Δ	both models in position band
`s_arrow`	Claude > GPT	0.503	1.000	-0.497	no
`s_convexity`	Claude > GPT	0.669	0.933	-0.264	no
`s_contrast`	GPT > Claude	1.000	0.630	+0.370	yes
`s_density`	GPT > Claude	1.000	0.814	+0.186	yes

Add `c_open_cplx` (Δ −0.276) for a third pair in the Claude direction, plus `s_size` or `s_symmetry` as a shared-focal positive control (both models 1.000) and Step 1's `noise` as a floor control.

Discard three of the eight that passed. `s_complexity` despite the largest Δ: its 20-gon is perceptually a circle, so the pair measures prototypicality and duplicates `proto`. `c_elong_sym` and `c_star_circle`: Claude's form consistency there is 0.07 and 0.13, meaning its identity choice is entirely positional — "Claude does not coordinate here" is not interpretable when Claude is not engaging with the stimulus at all.

The two directions are not the same kind of effect.

In the GPT > Claude pairs, Claude is genuinely tracking form, just less consistently (form consistency 0.80 and 0.93) — a *graded difference in concentration*. In the Claude > GPT pairs, GPT is partly falling back to position (form consistency 0.47–0.60) — a *difference in kind*. Both support the claim "these models have different focal points". Only `s_contrast` and `s_density` support the stronger claim "model X converges where model Y is genuinely indifferent", and both are the same direction. If that distinction matters, more candidates are needed in the Claude > GPT direction, screened for GPT staying near P(left) = 0.5 while failing to concentrate. `s_arrow` suggests semantic and canonical-orientation cues as the place to look.

6. Limitations

- **Winner's curse.** The screen values are n=30 estimates selected as the extremes of 30 comparisons and are therefore inflated. A confirmation run at n=60 on fresh trials is required before these numbers are used for anything but stimulus selection. At n=30 a match-rate CI is roughly ±0.15.
- **Ceiling.** 18 of 30 screened pairs were uninformative. A future battery needs stimuli calibrated to the middle of the range, not to obvious focal points.
- **Simulated dyads.** No two instances ever played each other. Match probability is computed from the marginal choice distribution, which is the correct population quantity for independent draws but is not a played game.
- **Two models, one settings profile.** Both models ran at low reasoning effort. Effort is known to change tool-use and deliberation behaviour and was not varied here.
- **No API-level seed.** Neither endpoint exposes one, so runs are not bit-reproducible; only the counterbalance allocation is seeded.

7. Files

All paths relative to `Name_game_experiment/`.

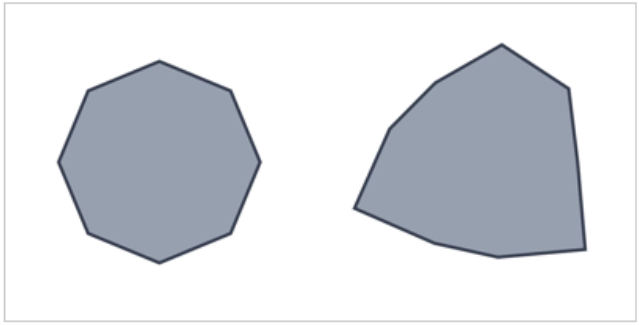
file	contents
`step1_form_choice.py`	Step 1 runner — prompt, settings, counterbalance
`step1_form_analysis.py`	Step 1 statistics and Figure 1
`step1_output/step1_form_calls_20260804_124918.csv`	600 call-level rows with full reasoning traces
`step1_output/step1_deliverables_20260804_124918/`	packaged raw data, summary, comparison, run config
`make_screen_stimuli.py`	generates the 30-pair battery and manifest
`step1b_screen.py`	screen runner; imports the prompt from Step 1
`step1b_screen_analysis.py`	selection criteria and diagnostics
`step1b_screen_figure.py`	Figure 2
`step1_output/step1b_screen_calls_20260804_153626.csv`	1,800 call-level rows
`step1_output/step1b_screen_selected_20260804_153626.csv`	per-pair criteria and crossover flags

Appendix — the 30-pair screening battery

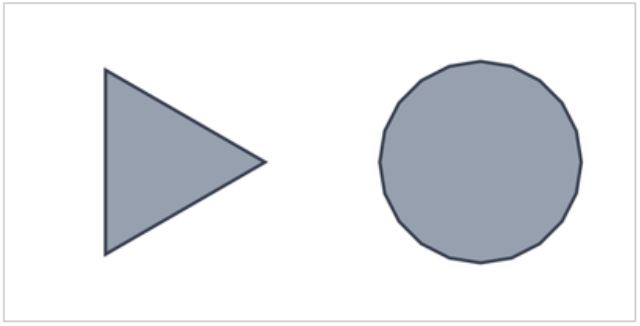
Each panel shows the _AB layout: identity 1 left, identity 2 right. Position is counterbalanced across trials.

Single-cue pairs (left panel = identity_1)

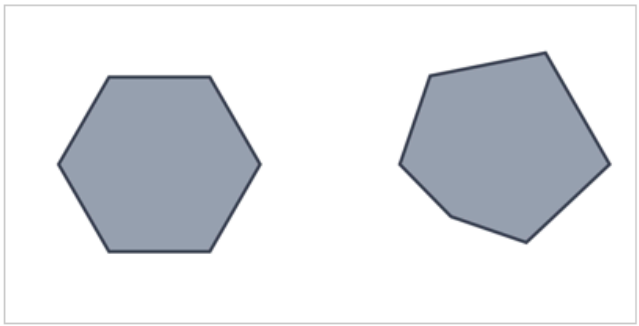
s_symmetry
symmetric | asymmetric — symmetry



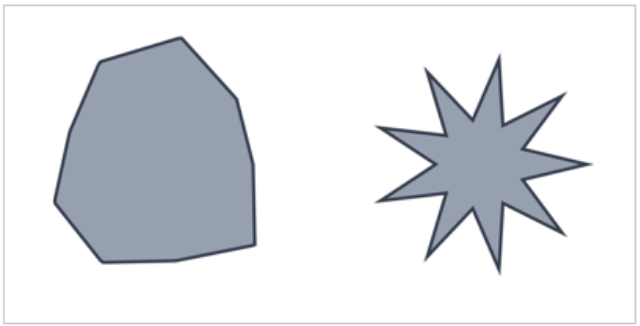
s_complexity
simple | complex — complexity



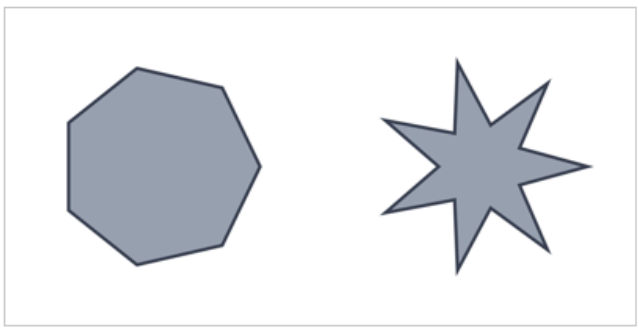
s_regularity
regular | irregular — regularity



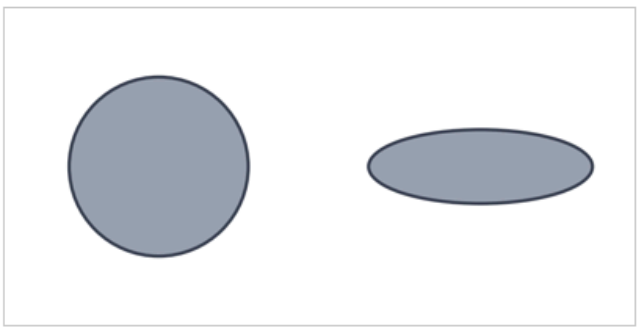
s_curvature
smooth | spiky — curvature



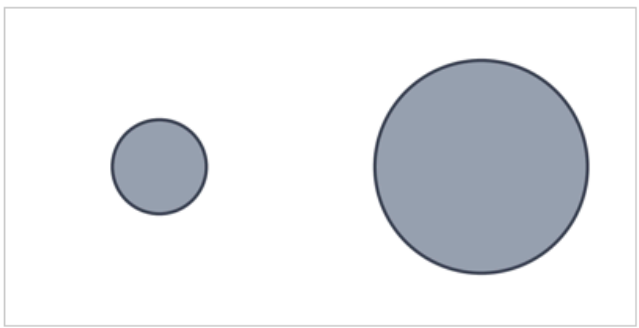
s_convexity
convex | concave — convexity



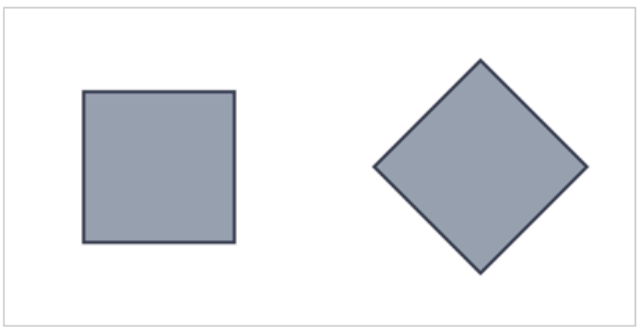
s_elongation
compact | elongated — elongation



s_size
small | large — size



s_tilt
upright | tilted — orientation

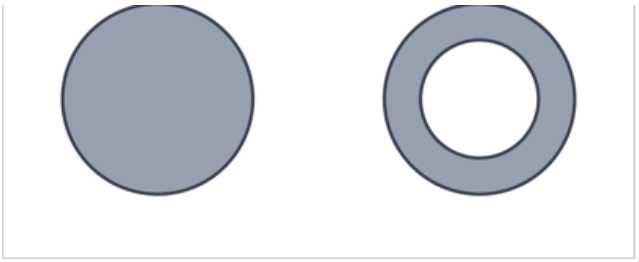


s_closure
closed | open — closure

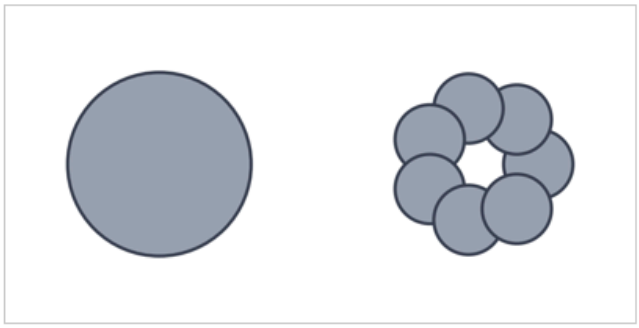
s_hole
solid | holed — hole



s_count
one | many numerosity



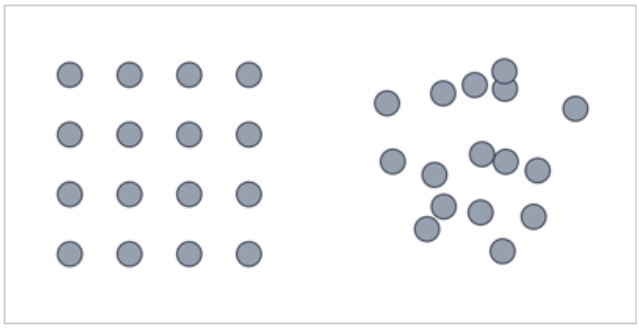
s_density
sparse | dense density



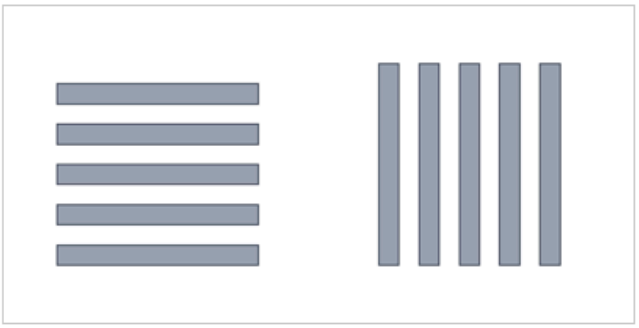
s_alignment
aligned | scattered alignment



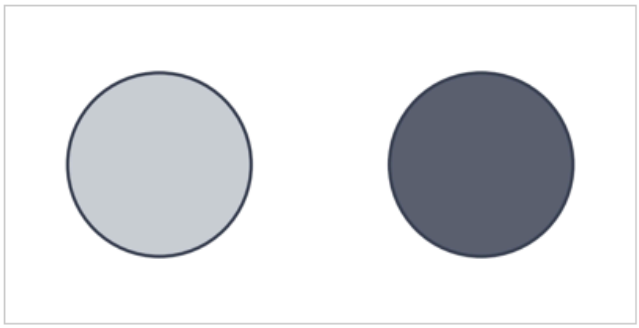
s_stripe_or
horizontal | vertical line orientation

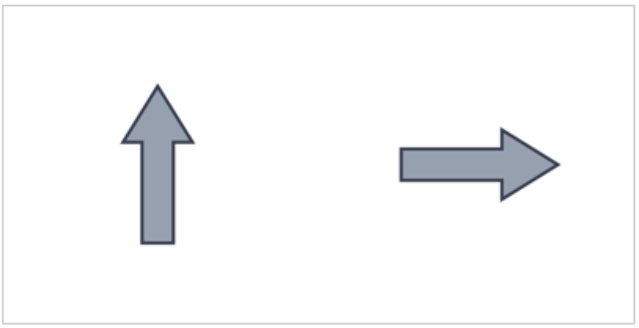


s_contrast
light | dark contrast



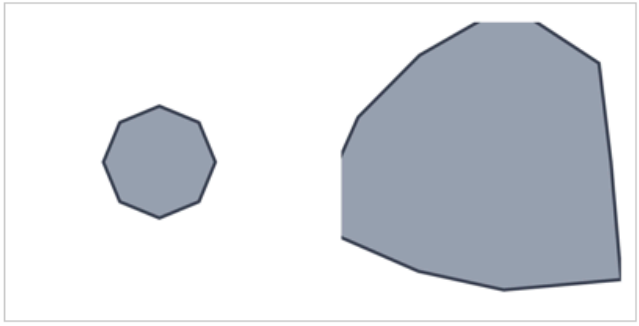
s_arrow
up | right semantic orientation



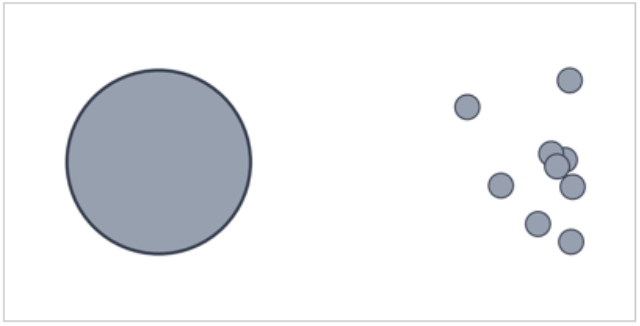


Competing-cue pairs (left panel = identity_1)

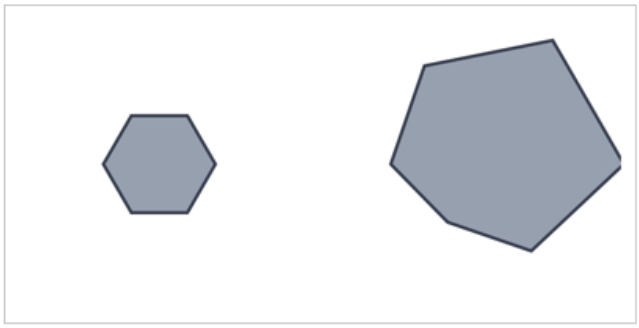
c_sym_size
sym_small | asym_large — symmetry vs size



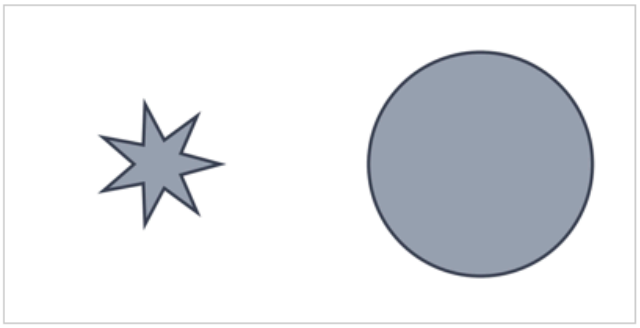
c_sym_count
sym_one | asym_many — symmetry vs numerosity



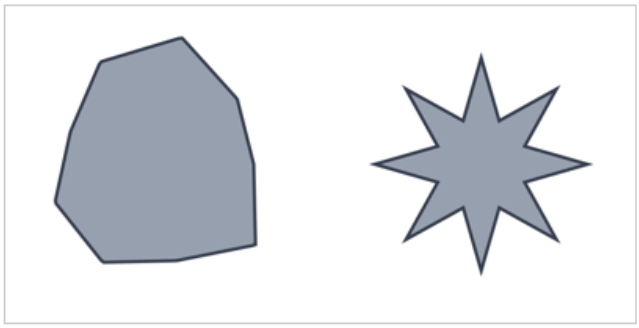
c_reg_size
reg_small | irreg_large — regularity vs size



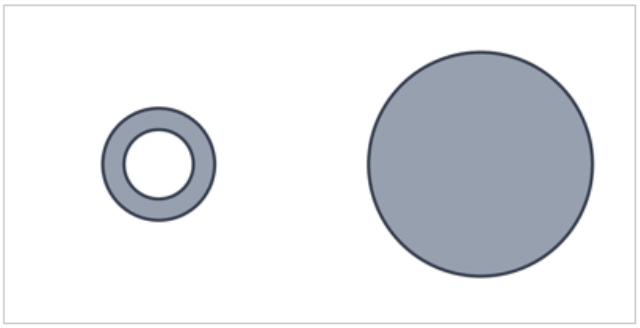
c_spiky_size
spiky_small | smooth_large — curvature vs size



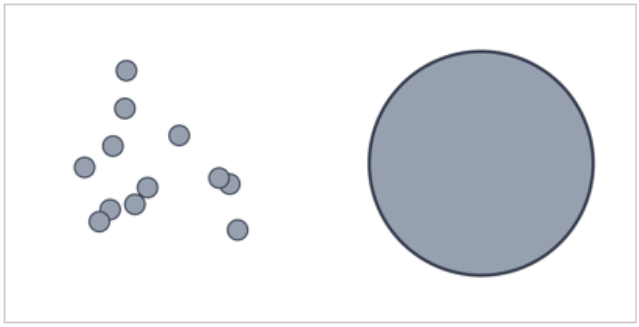
c_smooth_sym
smooth_asym | spiky_sym — curvature vs symmetry



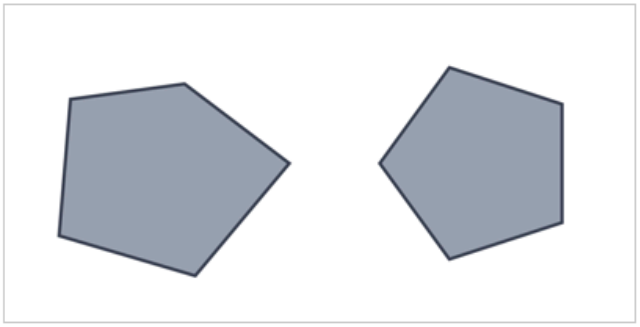
c_hole_size
holed_small | solid_large — hole vs size



c_count_size
many_small | one_large — numerosity vs size



c_tilt_reg
upright_irreg | tilted_reg — orientation vs regularity

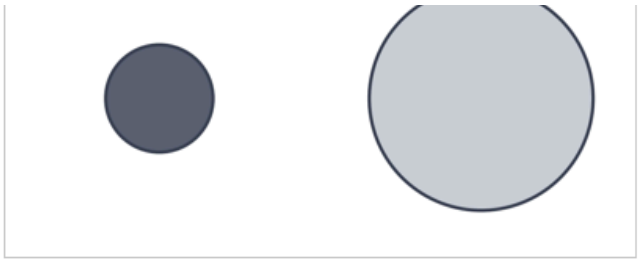


c_contr_size
dark_small | light_large — contrast vs size

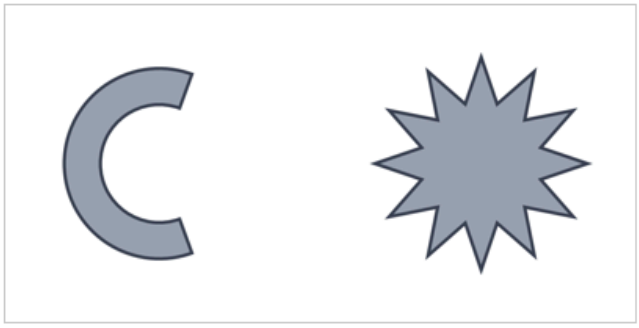


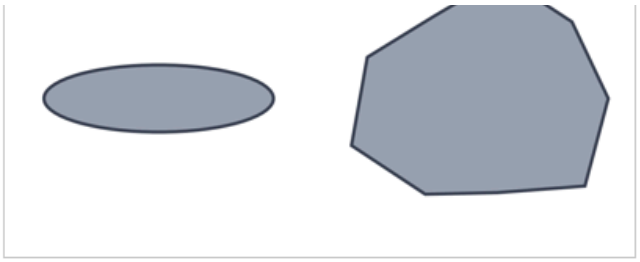
c_elong_sym
elong_sym | compact_asym — elongation vs symmetry





c_open_cplx
open_simple | closed_cplx — closure vs complexity

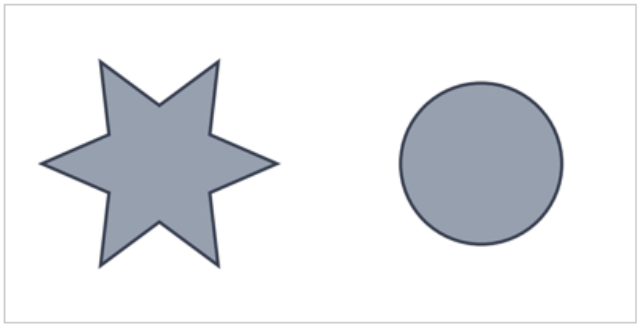




c_dens_align
dense_scatter | sparse_grid — density vs alignment



c_star_circle
star_eqarea | circle_eqarea — curvature vs area-matched



c_nested_big
nested_small | single_large — nestedness vs size

